

# Module 3 KPI Feature Engineering

## Methodology

This module focuses on transforming a cleaned military dataset into an **analytics-ready dataset** by engineering meaningful KPIs and enriching it with geopolitical metadata for visualization platforms like **Power BI and Tableau**.

### Step 1: Data Loading

- Imported the cleaned dataset (`military_cleaned.csv`) using **Pandas**
- Verified successful loading and dataset dimensions

### Step 2: Numeric Data Standardization

- Converted key quantitative columns (population, budgets, assets, power index) into numeric types
- Used error coercion to safely handle missing or malformed values

### Step 3: KPI Feature Engineering

Three high-impact KPIs were computed:

#### KPI 1: Power Index Rank Gap

- Computed a derived rank based on `Power_Index` (lower value = stronger military)
- Calculated the difference between computed rank and official rank
- Helps identify countries that are over-ranked or under-ranked

#### KPI 2: Assets per Capita

- Aggregated total military assets (aircraft, helicopters, tanks, naval fleet)
- Normalized total assets by population
- Enables fair comparison across countries with different population sizes

#### KPI 3: Budget-to-GDP Ratio

- Divided defense budget by purchasing power parity (GDP)
- Measures how much economic capacity is allocated to defense

### Step 4: Metadata Enrichment

- Added geopolitical context using mapping logic:
  - **Continent**
  - **Region**
  - **NATO Membership Flag**

- Filled missing mappings with "Other" to maintain dataset integrity
- Enables alliance-based and regional analysis

#### Step 5: Data Restructuring for Visualization

- **Wide Format:** One row per country with all metrics and KPIs
- **Long Format:** KPI-centric structure using `pd.melt()` for dynamic dashboards

#### Step 6: Export & Delivery

- Exported final output to `military_final.xlsx`
- Created two sheets:
  - `wide_format`
  - `long_format`
- Ensured direct compatibility with Power BI and Tableau

#### Insights

- **Power Index Rank Gap** reveals inconsistencies between perceived ranking and calculated military strength, helping identify strategically underrated or overrated nations.
- **Assets per Capita** provides a realistic measurement of force density, avoiding bias caused by large populations.
- **Budget-to-GDP Ratio** highlights national defense priorities relative to economic strength.
- NATO membership tagging enables **alliance-level military comparisons**.
- Region and continent enrichment supports **geopolitical storytelling** and regional power analysis.
- Providing both wide and long formats ensures **flexible KPI visualization**, ranking analysis, and filtering in BI tools.
- The engineered dataset significantly reduces preprocessing effort at the dashboard stage.

#### Libraries Used

| Library           | Purpose                                                                                  |
|-------------------|------------------------------------------------------------------------------------------|
| <b>pandas</b>     | Data loading, cleaning, transformation, KPI engineering, reshaping (wide & long formats) |
| <b>xlsxwriter</b> | Exporting analytics-ready Excel files with multiple sheets                               |
| <b>os</b>         | File and directory handling                                                              |